

Exploring Robustness and Redundancy in Distilled Encoder – Decoder Models

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Abstract—Recent advances in model compression have produced compact encoder-decoder models that rival much larger decoder-only systems in instruction-following tasks while requiring significantly fewer computational resources. However, while current literature heavily emphasizes the optimization process itself, understanding how the internal components of these compressed systems contribute to final performance is essential for identifying architectural redundancies and ensuring robust deployment.

In this work, we analyze compact T5 and Flan-T5 variants to study how their internal components behave under targeted modifications before and after the distillation process. We target two main distillation approaches: Semantic Knowledge Distillation and MiniPLM, and then analyze how these methods affect the students’ internal mechanics. To analyze these models, we introduce a systematic ablation framework that evaluates multiple architectural dimensions. This framework includes structural changes affecting encoder and decoder depth, targeted interventions on feed-forward and attention modules, and robustness-oriented perturbations applied directly to internal representations.

We evaluate instruction-following examples from *Dolly-15k* and *Self-Instruct*. To capture both strict alignment and conceptual preservation, we evaluate model outputs using a combination of traditional lexical overlap metrics, specifically ROUGE-L and METEOR, alongside embedding-based semantic similarity powered by Qwen. This multi-faceted approach enables us to measure exact text degradation while accurately tracking semantic consistency under each ablation setting.

Our study provides insights into the robustness and architectural redundancies of distilled encoder-decoder models under constrained computational budgets. More broadly, we offer a lightweight framework to analyze compact language models, guiding future efforts in model simplification and efficient deployment.

Index Terms—Knowledge Distillation, Encoder-Decoder Models, T5, Flan-T5, Ablation Study, Model Compression, Semantic Textual Similarity, Attention Mechanisms.

I. INTRODUCTION

While Large Language Models (LLMs) excel at open-ended text generation and instruction following, their substantial

memory and latency requirements often restrict their deployment under constrained hardware budgets. To bridge this gap, model compression and knowledge distillation are frequently leveraged to produce efficient, compact alternatives, often built on encoder-decoder architectures like T5 [1] and Flan-T5 [2]. However, while these optimized systems are routinely evaluated on overall task accuracy, the internal dynamics, redundancies, and robustness of their constituent components post-compression has received far less systematic analysis.

This paper investigates the behavior of encoder-decoder language models under systematic architectural perturbations. Our broader research initiative consists of two complementary tracks: first, the distillation of a decoder-only system into a compact encoder-decoder student model; and second, which forms another focus of this work, a comprehensive empirical analysis of these architectures under targeted ablations. By evaluating both base and distilled model variants, we isolate the architectural sensitivity of these models, treating the distillation procedure itself as a background context while centering our core contribution on the dynamics of post-distillation model degradation.

We evaluate T5-large on the *Dolly-15k* dataset and Flan-T5-large on the *Self-Instruct* dataset, comparing model behavior before and after targeted interventions. Our experimental framework operates across three distinct levels of granularity. First, we perform block-level dropping to evaluate structural necessity, testing individual encoder and decoder blocks, cumulative sequences (from both the beginning and the end of the networks), alternating odd/even layers, and complete feed-forward sublayers. Second, we apply threshold-based masking to selectively mute attention heads (encompassing self-attention and decoder cross-attention) and feed-forward modules at varying ratios. Third, we inject zero-mean Gaussian noise at increasing standard deviations directly into internal block-level and feed-forward representations to assess model resilience under continuous perturbations rather than

discrete structural removal.

Our central hypothesis is that while distilled encoder-decoder models exhibit measurable structural redundancy, this redundancy is non-uniformly distributed across their internal components. Specifically, we expect encoder layers, decoder self-attention, decoder cross-attention, and feed-forward blocks to display distinct degradation profiles under ablation. By subjecting both intact base models and distilled variants to the same perturbation protocol, we aim to determine whether distillation preserves structural robustness globally or whether it concentrates functional dependence into a smaller, more fragile subset of layers and attention heads.

We evaluate degradation using both embedding-based semantic similarity and lexical metrics such as ROUGE-L and METEOR.

The main contributions of this paper are as follows:

- We propose a structured ablation framework designed specifically for encoder-decoder language models, incorporating block-level dropping, feed-forward removal, multi-granularity attention-head masking, and Gaussian noise injection.
- We conduct a comparative analysis of intact and distilled encoder-decoder models under identical perturbation settings, analyzing how distillation affects the overall distribution of architectural redundancy.
- We evaluate model robustness on *Dolly-15k* and *Self-Instruct*, providing experimental coverage of both human-written and synthetically generated instruction-following data.
- We combine modern embedding-based semantic similarity with ROUGE-L and METEOR, establishing an evaluation strategy that simultaneously measures conceptual preservation and lexical fidelity under structural degradation.

II. RELATED WORK

A. Architectures and Training Techniques

The foundation of our work rests on the text-to-text transfer framework introduced by T5 [1] and the instruction-finetuning paradigms formalized in Flan-T5 [2]. A noteworthy approach in this domain is the Mixture-of-Denoisers (MoD) objective utilized in UL2 [3], which dynamically alters corruption spans and masking rates during pretraining. In terms of creating compact variants, knowledge distillation techniques, pioneered by DistilBERT [4] and TinyBERT [5], have demonstrated that smaller student models can replicate teacher behaviors by matching internal representations. Furthermore, advanced distillation frameworks like MiniLM [6] explicitly transfer self-attention relations, showing that localized attention maps hold highly compressible structural knowledge.

B. Relation of Prior Works and Current Shortcomings

These prior works broadly represent two parallel evolutionary paths: the scaling of highly capable encoder-decoder architectures (such as T5 and UL2) and the development of

compression paradigms (such as TinyBERT and MiniLM) originally tailored for encoder-only models. However, a major limitation in the field is that model compression is primarily evaluated using holistic, downstream end-task scores. While structural pruning methods, such as LayerDrop [7] for depth reduction and the unified NASH framework [8] for encoder-decoder models, show that these networks are significantly over-parameterized, these works focus heavily on accelerating inference rather than mapping internal semantic fragility. Similarly, Michel et al. [9] and Voita et al. [10] demonstrate that many attention heads can be removed with limited degradation and that only a subset of heads appears task-critical. Our work addresses this gap by using systematic ablations not to construct a new compressed model, but to analyze the structural sensitivity and robustness of existing base and distilled encoder-decoder architectures.

C. Datasets Used and Rationale

To evaluate instruction-following robustness, we use two datasets: *Dolly-15k* [11] and *Self-Instruct* [12].

Instruction-following robustness has recently been studied across both human-authored and synthetic instruction corpora [13], [14].

D. Benchmarks, Evaluation Methods, and Limitations

Traditional generation benchmarks heavily rely on rigid n -gram lexical overlap metrics like ROUGE or BLEU. However, as demonstrated by Zhang et al. [15] and Sellam et al. [16], such metrics penalize valid semantic paraphrasing, making them less suitable for open-ended generation tasks. Therefore, our primary evaluation strategy leverages semantic textual similarity via the modern Qwen3-Embedding model to calculate the semantic delta (Δ) between the baseline outputs and the ablated student outputs.

We acknowledge that while embedding-based similarity generally captures semantic similarity more effectively than purely lexical overlap metrics, it is not without limitations. Recent studies, including ALIGN-SIM [17] and Nikiema et al. [18], indicate that embedding spaces can occasionally struggle to distinguish between semantic opposites (such as negations) or small linguistic transformations.

III. EXPERIMENTAL SETUP

A. Datasets

To evaluate the robustness of instruction-following capabilities, we utilize two primary datasets that represent both human-authored and synthetically generated instructions.

1) *Databricks Dolly-15k*: The *Dolly-15k* dataset [11] consists of approximately 15,000 high-quality, human-generated instruction-following records. These examples span a diverse range of task categories, including open-domain question answering, summarization, brainstorming, and information extraction. For our experiments, the input prompt x is formulated by concatenating the instruction and context fields:

$$x = [\text{Instruction}] \parallel [\text{Context}] \quad (1)$$

In instances where no context is provided, the instruction field is used as the sole input.

2) *Self-Instruct*: The *Self-Instruct* dataset [12] is a large-scale corpus of instructions generated through a semi-automated bootstrapping process using vanilla language models. This dataset provides broader task coverage and linguistic variety compared to human-only datasets, allowing us to test model resilience across a wider distribution of synthetic prompts. Evaluation is performed on the standard test split.

B. Baseline and Distilled Models

We focus our analysis on two foundational encoder-decoder architectures:

- **T5**: The original Text-to-Text Transfer Transformer [1], which serves as our baseline for general language modeling and transfer learning.
- **Flan-T5**: An instruction-tuned variant of T5 [2] utilized to determine whether instruction finetuning alters the structural redundancy or robustness of the network.

C. Distillation Setup

Our distillation setup bridges the Qwen3.5-4B [21], a larger decoder-only teacher model, and the smaller Encoder-Decoder students T5-large and Flan-T5-large without relying on complex, online tensor alignment. To overcome the inherent architectural and tokenizer mismatches, we leverage two primary methods. First, we employ Symbolic Knowledge Distillation [19], extracting the teacher’s implicit capabilities by having it generate an explicit, high-quality synthetic dataset. Second, we integrate the MiniPLM [20] framework to perform offline Difference Sampling. By scoring the synthetic dataset using both the teacher and a smaller reference model from the same family (Qwen3.5-0.8B [21]), we filter out redundant or trivial examples. This purely data-curation approach ensures the T5-based students train only on the most sophisticated, high-value examples while entirely sidestepping cross-architecture probability math.

D. Model Configuration and Decoding

All models, specifically *T5-large* and *Flan-T5-large* (approx. 770M parameters), are evaluated using deterministic greedy decoding to ensure reproducibility across ablation trials. We fix the generation parameters to `do_sample=False` and `num_beams=1`.

To standardize the computational overhead across the extensive permutations of ablations, we impose a maximum input sequence length of 128 tokens. For *T5-large*, we enforced a maximum generation limit of 256 tokens per output instance, while for *Flan-T5-large*, it was set to 128. Experiments were conducted using the Hugging Face *Transformers* library in a PyTorch environment. Given the iterative nature of the ablation framework, each perturbation setting is evaluated on a fixed subset of 128 randomly selected examples from the respective dataset test splits to provide a consistent comparison baseline.

IV. ABLATION FRAMEWORK AND EXPERIMENTAL PROTOCOL

To systematically investigate the structural robustness of the *T5-large* and *Flan-T5-large* models, we propose a three-part ablation framework. Each experiment is conducted on a fixed subset of $N = 128$ samples from the *Dolly-15k* and *Self-Instruct* test splits. All perturbations are measured against a *baseline_no_drop* configuration using greedy decoding (`do_sample=False`).

A. Part 1: Structural and Feed-Forward Dropping

We first evaluate the impact of discrete component removal to identify layer-wise redundancy. This track includes:

- **Single Block Dropping**: Independent removal of encoder block i or decoder block i .
- **Cumulative Dropping**: Sequential removal of blocks starting from the shallow layers ($L_0 \dots L_i$) or deep layers ($L_n \dots L_i$).
- **Layer Parity Ablation**: Targeted removal of all even-indexed, all odd-indexed, or both sets of blocks across the encoder, decoder, or the entire unified architecture.
- **Feed-Forward (FF) Removal**: Selective dropping of the feed-forward sub-layers using the same parity-based logic (even, odd, or global removal).

B. Part 2: Threshold-based Masking

To observe the effect of partial information loss, we apply masking to the internal attention and feed-forward mechanisms using a range of thresholds $\tau \in \{0.1, 0.25, 0.5, 0.75, 0.8\}$.

- **Attention Masking**: We selectively mask attention heads in the encoder self-attention, decoder self-attention, decoder cross-attention, and combinations of these components.
- **Feed-Forward Masking**: Masking is applied to the FFN sub-layers at increasing ratios across the encoder and decoder to determine the "breakpoint" of semantic coherence.

C. Part 3: Stochastic Gaussian Perturbation

Unlike discrete removal, we assess model resilience under continuous noise by injecting zero-mean Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$ directly into the block representations and feed-forward layers. We test the standard deviation range $\sigma \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1.0\}$. This allows us to map the stability of the hidden state space under increasing variance.

D. Evaluation Metrics

The results of these ablations are quantified via the following metrics, visualized in Section V:

- 1) **Semantic Similarity** (μ_{sim}): The primary metric, calculated as the cosine similarity between the embeddings of the generated and reference text using the *Qwen3-Embedding-0.6B* model.
- 2) **Hard Semantic Accuracy**: The fraction of samples where $\mu_{sim} > 0.5$ serves as a proxy for conceptual preservation.

- 3) **Lexical Metrics:** Standardized ROUGE scores (1, 2, L, Lsum) and METEOR are utilized to track exact-match degradation and linguistic fluency.

V. RESULTS AND DISCUSSION

A. Distillation Results

Table I presents the embedding similarity scores between teacher-generated and student-generated outputs. These distillation scores and the ablation experiments are reported using Qwen3-Embedding-0.6B to compare generated outputs against dataset references. The results indicate a pronounced performance disparity between the two distillation strategies. Semantic Knowledge Distillation consistently yields stronger teacher-student embedding alignment across both the Dolly and Self-Instruct datasets, peaking at a similarity of 0.8754 with the Flan-T5-Large student.

Architectural initialization also appears to influence the effectiveness of MiniPLM. While the instruction-tuned Flan-T5-Large initialization benefits the Semantic method, it performs worse under MiniPLM compared with the base T5-Large architecture. For example, on Dolly, MiniPLM decreases from 0.6651 with T5-Large to 0.5489 with Flan-T5-Large. This suggests that MiniPLM’s difference-sampling methodology, which filters the dataset toward higher-divergence examples, may interact less favorably with the instruction-tuned initialization of Flan-T5 than with the base T5 architecture.

Dataset	Distillation Method	T5-Large	Flan-T5-Large
Dolly	Semantic	0.8341	0.8754
	MiniPLM	0.6651	0.5489
Self Instruct	Semantic	0.8232	0.8276
	MiniPLM	0.7235	0.6183

TABLE I
EMBEDDING SIMILARITY SCORES FOR SEMANTIC KD AND MINIPLM
DISTILLATION, EVALUATED WITH QWEN3.5-0.8B.

B. T5 and Students Comparison

We analyze the ablation behavior of *T5-Large* and its two distilled variants on the *Dolly-15k* dataset. The two student models correspond to Semantic Knowledge Distillation and MiniPLM distillation. Since no separate baseline file is available for the student ablations, their baselines are inferred from the near-zero Gaussian perturbation setting at $\sigma = 0.01$. Under this approximation, the MiniPLM student obtains an inferred baseline similarity of 0.6703, with ROUGE-L 0.2471 and METEOR 0.3025, while the Semantic student obtains an inferred baseline similarity of 0.5505, with ROUGE-L 0.1556 and METEOR 0.2594. These inferred values indicate that the MiniPLM student starts from a substantially stronger semantic and lexical baseline than the Semantic distillation student.

1) *Base T5 Ablation Results:* The base *T5-Large* model is generally robust to isolated encoder block removal. The most sensitive encoder block is block 0, whose removal produces the largest encoder-side degradation, with $\Delta_{\text{sim}} = -0.0293$, $\Delta_{\text{ROUGE-L}} = -0.0313$, and $\Delta_{\text{METEOR}} = -0.0268$.

However, this degradation remains moderate, and 14 out of the 24 encoder blocks slightly improve similarity when removed individually. This suggests that the encoder stack contains substantial block-level redundancy under isolated removal (Fig. 1).

The decoder also exhibits robustness to most isolated block removals, but block 0 is a clear exception. Removing decoder block 0 causes the largest decoder-side degradation, with $\Delta_{\text{sim}} = -0.137$. Other decoder blocks have much smaller individual effects. Therefore, the base *T5-Large* model differs from the Flan-T5 behavior observed later: its dominant decoder bottleneck appears at the start of the decoder rather than at an intermediate decoder block (Fig. 2).

Cumulative dropping reveals that the apparent redundancy under isolated removal does not extend to progressive structural truncation. Removing blocks from the beginning of the encoder is catastrophic: after removing only the first two encoder blocks, similarity decreases to approximately 0.28, while ROUGE and METEOR approach zero. Decoder-side cumulative removal from the beginning is also highly destructive, with one or two removed blocks already producing major degradation. In contrast, removing blocks from the end produces a slower decline, with approximately 5-10 terminal blocks removable before severe collapse. This indicates that early encoder and decoder blocks are essential for constructing and routing representations, while later blocks are more redundant (Fig. 3).

Odd/even dropping reveals a strong parity asymmetry. Odd-block removal has almost no effect: $\Delta_{\text{enc}} = +0.005$, $\Delta_{\text{dec}} = +0.009$, and $\Delta_{\text{all}} = +0.001$. Even-block removal is substantially more harmful, especially in the decoder: $\Delta_{\text{dec}} = -0.214$ and $\Delta_{\text{all}} = -0.288$, while the encoder drop is milder at $\Delta_{\text{enc}} = -0.050$. Thus, decoder performance depends more strongly on even-indexed blocks (Fig. 4).

Feed-forward dropping shows a similar structured-removal effect. Removing even encoder feed-forward sublayers increases similarity to 0.647, above the 0.611 base similarity. This suggests that some encoder feed-forward transformations may be non-essential or mildly harmful under the current evaluation metric. However, removing all feed-forward sublayers, or all decoder feed-forward sublayers, causes collapse, with similarity in the 0.21–0.30 range and near-zero ROUGE-L. Therefore, while selected feed-forward sublayers are redundant, the feed-forward pathway as a whole remains critical (Fig. 5).

Attention masking is far more destructive than isolated block dropping. At only 10% masking, similarity drops from the 0.611 baseline to approximately 0.30, and remains in the 0.22–0.36 range up to 75% masking. The most destructive configuration is joint encoder-decoder masking of self-attention and cross-attention, which reaches a minimum similarity of 0.217 at 75% masking. Decoder cross-attention is the most damaging individual attention component. These results show that random corruption of active attention computations is substantially more harmful than clean removal of complete blocks (Fig. 6).

Feed-forward masking is similarly destructive. While structured feed-forward dropping can preserve similarity above 0.6 in selected cases, random feed-forward masking reduces similarity to approximately 0.21–0.44 even at low masking ratios. Encoder feed-forward masking saturates almost immediately, remaining near 0.21 across masking ratios from 10% to 80%. This supports the interpretation that clean architectural bypassing is more survivable than partial corruption: when an entire sublayer is removed, residual connections can still transmit information, whereas random masking leaves a damaged computation active.

2) *Base T5 versus Semantic Distillation Student*: The Semantic Knowledge Distillation student starts from a weaker inferred baseline than the MiniPLM student, with similarity 0.5505, ROUGE-L 0.1556, and METEOR 0.2594. Under single-block dropping, the Semantic student shows a strong decoder-side dependence on block 0. Fig. 7 shows that removing decoder block 0 causes the largest collapse, with the similarity drop reaching approximately -0.30 , while most other decoder blocks remain close to zero change. This mirrors the base T5 behavior, where decoder block 0 is also the dominant isolated decoder bottleneck.

On the encoder side, the Semantic Knowledge Distillation student behaves differently from the base model. Removing encoder block 0 slightly improves semantic similarity, while reducing METEOR, indicating a disagreement between embedding-level semantic similarity and lexical quality. Most remaining encoder blocks produce only small positive or negative changes. This suggests that Semantic Knowledge Distillation preserves some encoder redundancy, but it redistributes sensitivity away from the exact base-model pattern (Fig. 8).

Cumulative dropping remains destructive for the Semantic student. Removing blocks from the beginning causes a rapid collapse in similarity and lexical metrics, while removing blocks from the end produces a more gradual decline. This reinforces the conclusion that the early encoder and decoder blocks are structurally important even after semantic distillation (Fig. 9).

3) *Base T5 versus MiniPLM Student*: The MiniPLM student obtains a stronger inferred baseline than the Semantic student, with similarity 0.6703, ROUGE-L 0.2471, and METEOR 0.3025. Under single decoder block dropping, MiniPLM again exhibits a strong dependence on decoder block 0. Removing decoder block 0 causes the largest similarity drop, approximately -0.28 , while most other decoder blocks produce changes close to zero. This indicates that MiniPLM preserves the same dominant decoder-side bottleneck observed in the base T5 model (Fig. 7).

The encoder behavior of MiniPLM is more evenly distributed. Some encoder blocks improve performance when removed, while others cause moderate degradation. The largest encoder-side drops occur near the final encoder blocks, suggesting that MiniPLM assigns greater importance to terminal encoder representations than the base model. This differs from the base model, where encoder block 0 is the most sensitive

isolated encoder block (Figs. 8 and 11).

Cumulative dropping shows that MiniPLM is robust to isolated block removal but fragile under progressive truncation. Removing encoder or decoder blocks from the beginning causes rapid degradation, while removal from the end is more gradual. This pattern indicates that MiniPLM preserves local redundancy but remains dependent on the early-stage processing pipeline (Fig. 12).

Overall, the T5 experiments show that MiniPLM preserves a stronger baseline and a more teacher-like decoder bottleneck than Semantic Knowledge Distillation. However, neither student removes the model’s structural fragility. Both distilled variants remain highly sensitive to decoder block 0 and to progressive removal from the beginning of the stack. Thus, for T5, distillation improves or preserves output-level fidelity without eliminating early-block dependence.

C. *Flan-T5 Ablation and Students Comparisons*

This subsection reports the ablation behavior of Flan-T5 on *Self-Instruct* and compares the intact Flan-T5 model with two distilled student variants: a semantic knowledge distillation student and a MiniPLM student. The base model obtains a baseline similarity of 0.6549. The semantic distillation student obtains a lower baseline similarity of 0.5896, while the MiniPLM student obtains a baseline similarity of 0.6572, slightly exceeding the base model under the same evaluation setting. Therefore, comparisons involving the semantic distillation student are interpreted primarily in terms of relative degradation under perturbation, whereas comparisons involving the MiniPLM student are interpreted both in terms of robustness and teacher-level baseline fidelity.

1) *Flan-T5 Base Ablation Results*: The Flan-T5 encoder is largely robust to single-block removal. Removing individual encoder blocks produces an average similarity change close to zero, with 14 out of 24 encoder blocks slightly improving similarity after removal. The most sensitive encoder block is block 23, whose removal causes a similarity drop of -0.0406 . Blocks 6, 7, and 8 are among the most removable encoder blocks, with block 7 producing the largest positive change of $+0.0115$.

By contrast, the decoder contains a clear critical block. Removing decoder block 6 causes the largest degradation, with $\Delta_{\text{sim}} = -0.2653$, $\Delta_{\text{ROUGE-L}} = -0.2675$, and $\Delta_{\text{METEOR}} = -0.1620$. No other single decoder block approaches this effect; the next largest decoder similarity drop is only -0.0134 . This indicates that Flan-T5’s ablation sensitivity is concentrated in a specific decoder block rather than distributed uniformly across the stack.

Odd/even dropping confirms that the decoder is more fragile than the encoder under structured removal. Dropping all decoder even layers reduces similarity to 0.3791, while dropping all decoder odd layers preserves substantially more similarity, reaching 0.6099. A similar but less selective degradation appears in the encoder, where dropping even layers yields 0.4077 similarity and dropping odd layers yields 0.4838. The

full even-layer drop is the most destructive parity intervention, reducing similarity to 0.3615.

Feed-forward interventions reveal a clear distinction between structured dropping and random masking. Dropping odd decoder feed-forward layers causes almost no degradation, reducing similarity from 0.6549 to 0.6526. In contrast, removing even decoder feed-forward layers lowers similarity to 0.3723. Removing all decoder feed-forward layers further reduces similarity to 0.3311 and nearly eliminates lexical overlap.

Random feed-forward masking is even more disruptive. At only 10% masking, similarity falls to 0.3728 for encoder masking, 0.4361 for decoder masking, and 0.3490 when masking is applied jointly.

Attention masking is less destructive than feed-forward masking, but it still reveals the importance of cross-attention. At 75% masking, decoder cross-attention reaches 0.5103 similarity, while masking decoder self-attention remains less harmful at 0.6344. The most damaging attention configuration is joint encoder-decoder self/cross masking, which reduces similarity to 0.3933.

2) *Flan-T5 Base versus Semantic Distillation Student*: The distilled student is more sensitive than the base model in the encoder. The base model’s worst encoder block is block 23, with $\Delta\text{sim} = -0.0406$. In the student, both encoder block 23 and encoder block 0 become more important, with drops of -0.0522 and -0.0516 , respectively. This suggests that distillation increases the importance of boundary encoder blocks.

The decoder comparison shows the opposite pattern for the most critical block. Both the base and student models identify decoder block 6 as the dominant failure point, but the student is less damaged by its removal. The base model drops by -0.2653 , while the student drops by -0.1828 . This suggests that distillation partially redistributes the dependency carried by decoder block 6, improving robustness to that specific structural deletion.

However, the student is less robust under attention masking. At 75% masking, decoder cross-attention similarity decreases from 0.5103 in the base model to 0.3790 in the student. Encoder self-attention also decreases from 0.5597 to 0.4839, and joint encoder-decoder self/cross masking decreases from 0.3933 to 0.3741. This indicates that while distillation can reduce sensitivity to one dominant decoder block, it also reduces attention-head redundancy.

Gaussian noise remains weakly informative in the student comparison. The student’s encoder-full noise setting remains close to baseline for small noise levels but decreases mildly at larger noise, reaching 0.5820 at $\sigma = 0.5$ and 0.5809 at $\sigma = 1.0$. Decoder-full noise remains almost unchanged, staying between 0.5892 and 0.5921. Therefore, the student shows a small encoder-side sensitivity to large Gaussian noise, but no meaningful decoder-side degradation.

Overall, Flan-T5’s most important ablation result is the concentration of structural sensitivity in decoder block 6. The distilled student preserves this vulnerability pattern, but with reduced severity for block deletion and increased sensitivity to

attention masking. This suggests that distillation does not uniformly improve robustness; instead, it shifts the failure mode from isolated layer dependence toward reduced redundancy in attention mechanisms.

3) *Flan-T5 Base versus MiniPLM Student*: We further compare the Flan-T5 base model against a second distilled variant trained using MiniPLM-style distillation. In contrast to the semantic knowledge distillation student, denoted as student v1, the MiniPLM student, denoted as student v2, preserves baseline performance almost entirely. The base model obtains a similarity score of 0.6549 and a hard semantic prediction score of 0.7734, while student v2 obtains 0.6572 similarity and 0.7812 hard semantic prediction. Student v2 also improves ROUGE-L from 0.2826 to 0.3007, while maintaining a competitive METEOR score of 0.1924.

This result indicates that MiniPLM distillation does not merely compress the model while preserving approximate output behavior; it also produces a student whose baseline semantic alignment is comparable to, and slightly higher than, the base model under the present evaluation setting. This differs from student v1, whose baseline similarity decreases to 0.5896 and whose hard semantic prediction score decreases to 0.5859.

Under single encoder block removal, student v2 follows the same broad pattern as the base model. Most encoder blocks produce only small changes in similarity, and the largest degradation remains concentrated near the final encoder block. This suggests that the MiniPLM student preserves the encoder redundancy pattern of the original model more closely than student v1.

The decoder analysis shows that student v2 preserves the same dominant failure point as the base model. In both models, removing decoder block 6 causes the largest performance degradation by a wide margin. Unlike student v1, where the severity of this failure was partially reduced, student v2 retains a sharper dependency on this decoder block. This indicates that MiniPLM distillation better preserves the structural behavior of the teacher and also retains its most important decoder bottleneck.

Cumulative layer dropping shows a clearer difference between isolated and progressive perturbations. Although student v2 is robust to many single-block removals, its hard prediction score decreases sharply when layers are removed cumulatively, especially from the beginning of the encoder or decoder. This suggests that MiniPLM preserves local layer redundancy but remains sensitive to large-scale structural truncation.

Additional MiniPLM ablations, including odd/even dropping, feed-forward dropping, feed-forward masking, and Gaussian noise injection, are reported in Appendix B. These experiments support the same conclusion: MiniPLM better preserves teacher-like behavior than semantic distillation, but it also retains the teacher’s structural weaknesses.

Attention masking reveals that student v2 remains close to the base model across most masking ratios. As in the base model, decoder cross-attention is among the most sensitive attention components, while decoder self-attention is more

robust. Compared with student v1, student v2 shows stronger preservation of attention robustness, suggesting that MiniPLM distillation retains more of the teacher’s attention-level redundancy.

4) Semantic Distillation versus MiniPLM Distillation:

Finally, we compare the two distilled students directly. Student v1 corresponds to semantic knowledge distillation, while student v2 corresponds to MiniPLM-style distillation. At the baseline level, student v2 is substantially closer to the base model. It improves similarity from 0.5896 in v1 to 0.6572, and hard semantic prediction from 0.5859 to 0.7812. ROUGE-L also improves from 0.2465 to 0.3007.

Under single-layer dropping, both students preserve the same dominant decoder sensitivity at block 6. However, student v2 more closely matches the base model’s degradation profile, while student v1 shows a weaker decoder-layer collapse and stronger irregularity across encoder layers. This suggests that MiniPLM better preserves teacher-like internal structure, whereas semantic distillation produces a student with altered robustness characteristics.

Cumulative dropping further separates the two students. Student v2 starts from a stronger baseline and generally tracks teacher-like behavior more closely, but it can still degrade sharply under progressive removal. Student v1 begins from a weaker baseline and shows less stable behavior across cumulative interventions. Therefore, MiniPLM improves baseline fidelity and teacher alignment, but it does not eliminate structural fragility under large-scale deletion.

Overall, the MiniPLM student provides the strongest distilled model among the evaluated variants. It preserves base-level semantic similarity, hard prediction accuracy, and lexical overlap more effectively than the semantic knowledge distillation student. However, its robustness profile remains teacher-like: it inherits the critical decoder block 6 dependence, the asymmetry between odd and even decoder components, and the high sensitivity of feed-forward masking. Thus, MiniPLM improves fidelity more than robustness; it compresses the teacher while retaining both its strengths and its structural weaknesses.

Additional Flan-T5 ablation figures are provided in Appendix B, including base-model block dropping, base-versus-student comparisons, MiniPLM-specific perturbations, and the cumulative comparison between semantic distillation and MiniPLM students.

VI. FUTURE WORK

Several extensions remain open for future work. First, the current analysis is limited to T5-Large and Flan-T5-Large on Dolly-15k and Self-Instruct. Extending the same ablation protocol to additional encoder-decoder families and larger instruction-following datasets would help determine whether the observed decoder bottlenecks are architecture-specific or more general.

Second, future experiments should evaluate whether the identified redundant components can be removed permanently through structured pruning followed by short retraining. Our

results show that some blocks and feed-forward sublayers can be ablated with limited degradation, but the current study does not optimize a compressed architecture after removal.

Third, a more fine-grained analysis of attention heads and feed-forward neurons could clarify whether the observed block-level fragility is caused by entire blocks or by a smaller subset of critical internal units. Methods such as head-level pruning, activation patching, and representation similarity analysis could provide a more precise localization of functional dependencies.

Finally, future work should include additional evaluation metrics and decoding settings. Since this study relies primarily on greedy decoding and similarity-based evaluation, testing beam search, sampling-based decoding, likelihood-based metrics, and human evaluation would provide a more complete picture of how structural perturbations affect generation quality.

VII. CONCLUSION

This work analyzed the robustness and redundancy of encoder-decoder language models before and after distillation. We evaluated *T5-Large* on *Dolly-15k* and *Flan-T5-Large* on *Self-Instruct*, applying a common ablation protocol based on block dropping, cumulative dropping, parity-based removal, attention and feed-forward masking, and Gaussian noise injection. Across both model families, the results show that robustness is highly non-uniform: some architectural components can be removed with little degradation, while others act as localized bottlenecks.

For *T5-Large*, isolated encoder and decoder block removal indicates substantial redundancy, but cumulative removal reveals a strong dependence on early blocks. Decoder block 0 is the most important single decoder block, and removing blocks from the beginning of either stack is far more destructive than removing blocks from the end. The T5 experiments also show a clear difference between structured removal and unstructured corruption. Dropping complete blocks or sublayers is often survivable, whereas attention and feed-forward masking rapidly collapse similarity, suggesting that corrupted active computations are more harmful than cleanly bypassed components.

For *Flan-T5-Large*, the main structural bottleneck is concentrated in decoder block 6. This differs from the T5 behavior, where the strongest decoder-side sensitivity appears at the beginning of the decoder. The Flan-T5 results also show that the encoder is comparatively redundant under isolated block removal, while decoder and feed-forward interventions produce sharper degradation. Attention masking further highlights the importance of decoder cross-attention, especially when comparing base and distilled variants.

The comparison between distillation methods shows that distillation does not preserve robustness uniformly. The semantic knowledge distillation student achieves weaker baseline fidelity and displays altered robustness behavior, particularly under attention masking. MiniPLM preserves baseline semantic similarity more effectively and remains closer to the

teacher’s robustness profile. However, this also means that MiniPLM inherits the teacher’s structural weaknesses, including the Flan-T5 decoder block 6 bottleneck and sensitivity to feed-forward corruption. Thus, MiniPLM improves fidelity more than it improves robustness.

Overall, our findings suggest that compact encoder-decoder models contain exploitable redundancy, but this redundancy is structured rather than uniform. Distillation can preserve output-level behavior while changing internal sensitivity, and stronger baseline performance does not necessarily imply greater robustness. Future work should extend this analysis to larger model families, additional instruction-following datasets, and more targeted interventions such as head-level pruning, activation patching, and layer-wise retraining after ablation.

The Project’s full code can be accessed at: <https://github.com/Razvan48/Project-LLMs-for-NLP>

APPENDIX A T5 ABLATION FIGURES

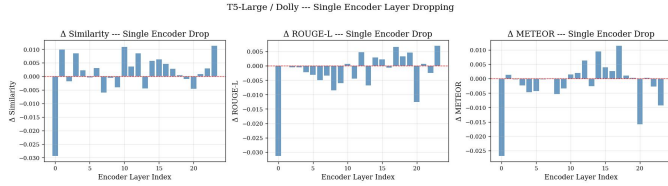


Fig. 1. Single encoder block dropping for base T5-Large.

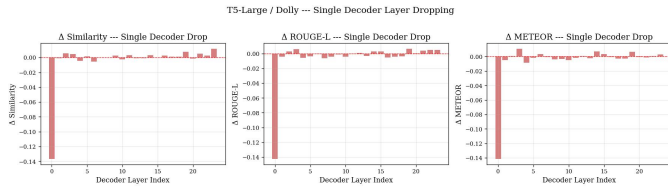


Fig. 2. Single decoder block dropping for base T5-Large.

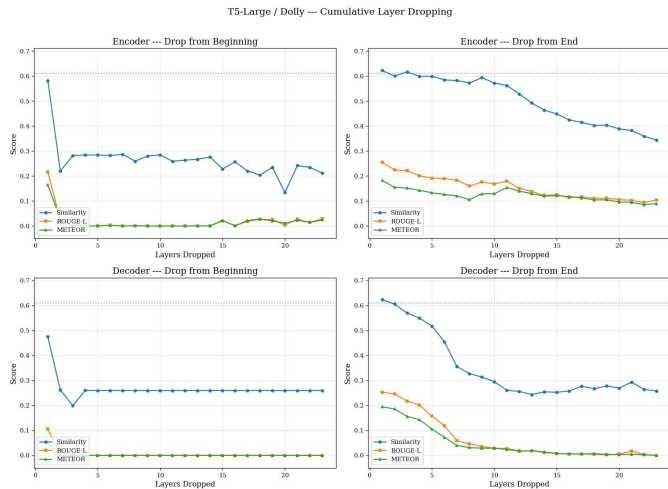


Fig. 3. Cumulative encoder and decoder block dropping for base T5-Large.

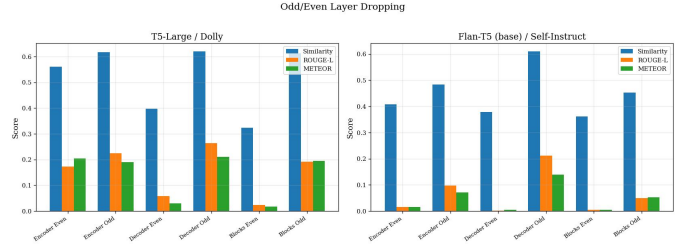


Fig. 4. Odd/even block dropping for base T5-Large.

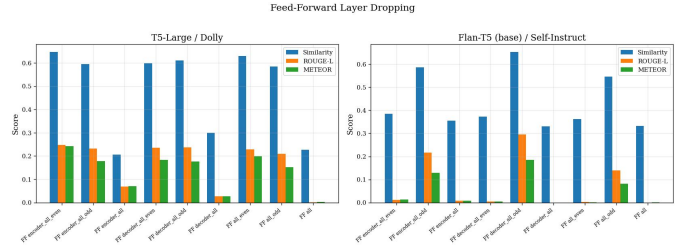


Fig. 5. Feed-forward sublayer dropping for base T5-Large.

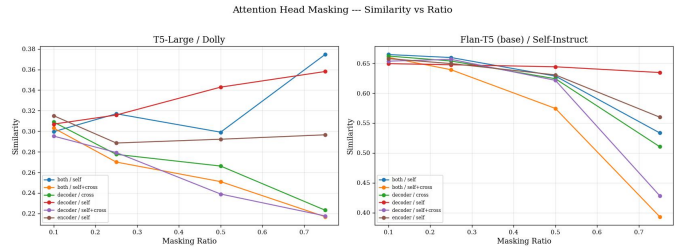


Fig. 6. Attention masking for base T5-Large.

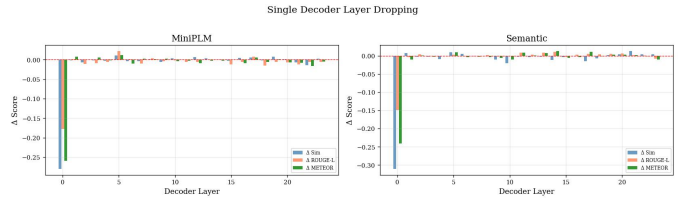


Fig. 7. Single decoder block dropping for the T5 MiniPLM and Semantic students.

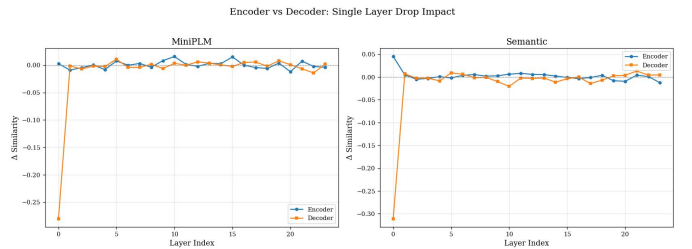


Fig. 8. Encoder-decoder sensitivity under single block dropping for the T5 students.

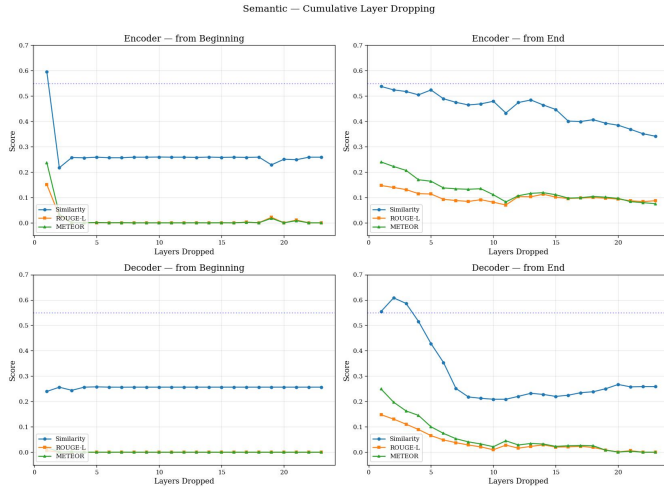


Fig. 9. Cumulative block dropping for the T5 Semantic student.

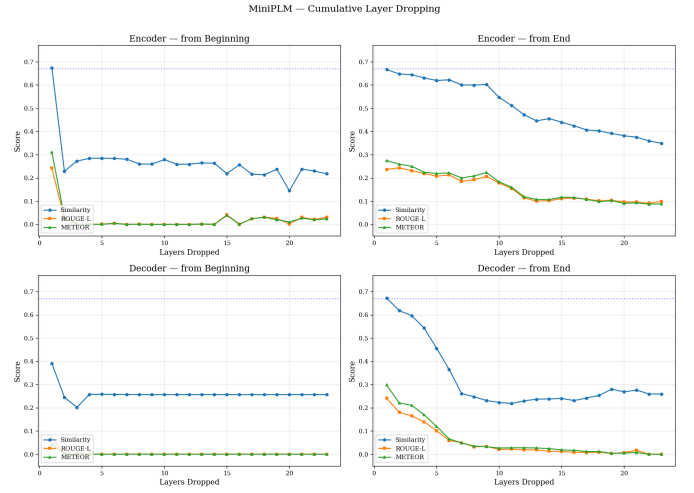


Fig. 12. Cumulative block dropping for the T5 MiniPLM student.

APPENDIX B FLAN-T5 ABLATION FIGURES

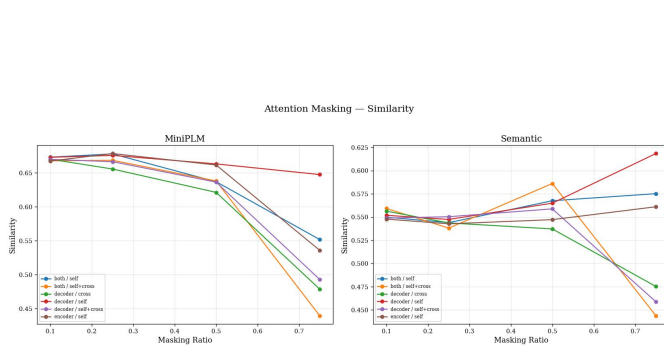


Fig. 10. Attention masking similarity for the T5 MiniPLM and Semantic students.

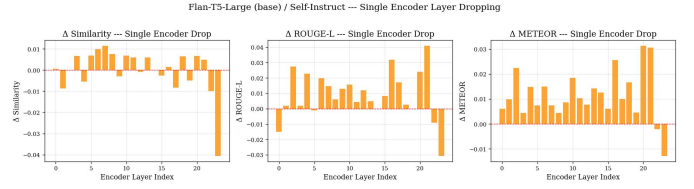


Fig. 13. Single encoder block dropping for the Flan-T5-Large base model on Self-Instruct.

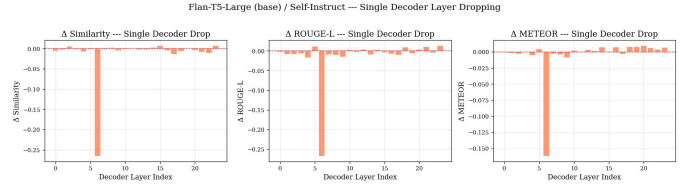


Fig. 14. Single decoder block dropping for the Flan-T5-Large base model on Self-Instruct.

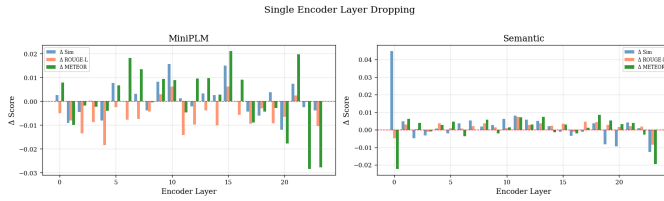


Fig. 11. Single encoder block dropping for the T5 MiniPLM and Semantic students.

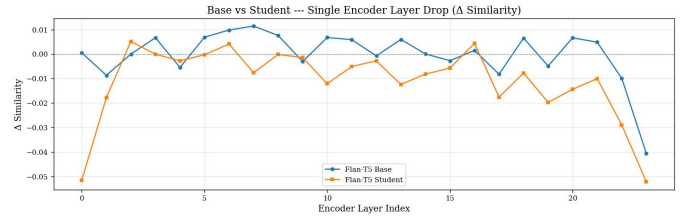


Fig. 15. Comparison between Flan-T5-Large base and distilled student under single encoder layer dropping.

Flan-T5 Base vs Student --- Gaussian Noise Robustness

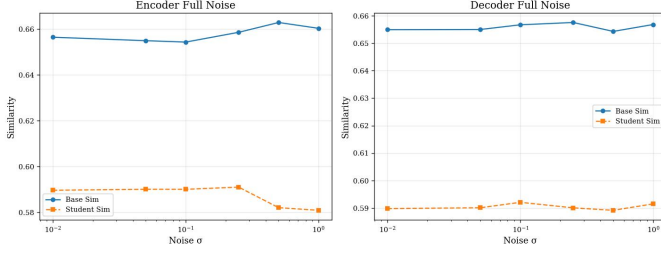


Fig. 16. Comparison between Flan-T5-Large base and distilled student under Gaussian noise injection.

Base vs v2 --- Encoder Drop

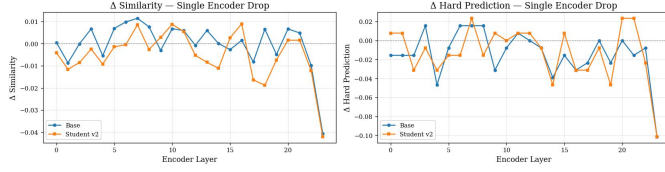


Fig. 17. Comparison between Flan-T5-Large base and MiniPLM student under single encoder block dropping.

Base vs v2 --- Cumulative Layer Dropping

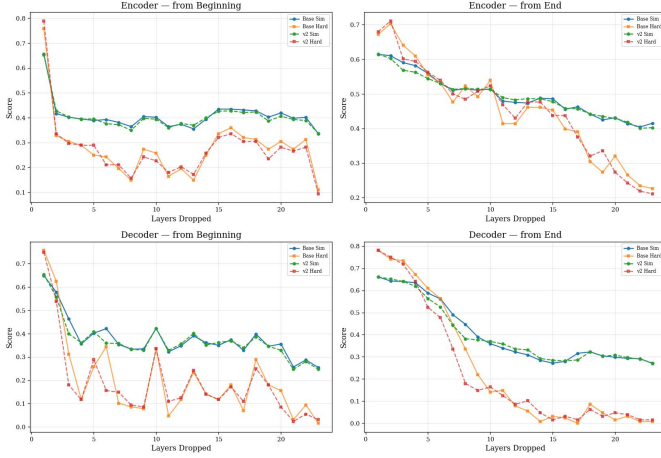


Fig. 18. Comparison between Flan-T5-Large base and MiniPLM student under cumulative encoder and decoder layer dropping.

Base vs v2 --- Odd/Even Dropping

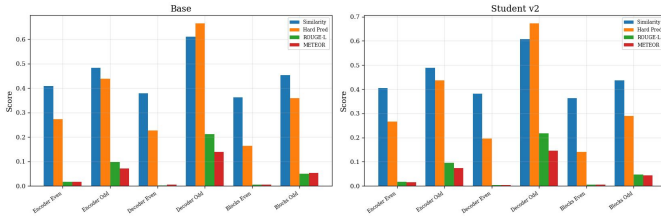


Fig. 19. Comparison between Flan-T5-Large base and MiniPLM student under odd/even layer dropping.

Base vs v2 --- FF Dropping

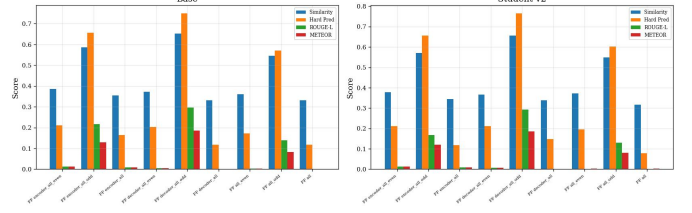


Fig. 20. Comparison between Flan-T5-Large base and MiniPLM student under feed-forward layer dropping.

Base vs v2 --- FF Masking

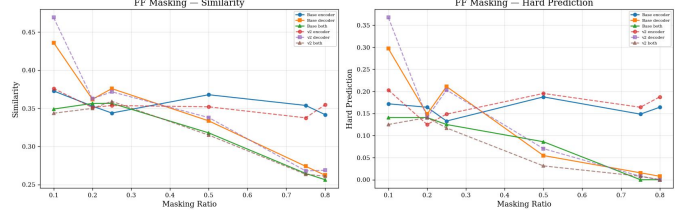


Fig. 21. Comparison between Flan-T5-Large base and MiniPLM student under feed-forward masking.

Base vs v2 --- Gaussian Noise

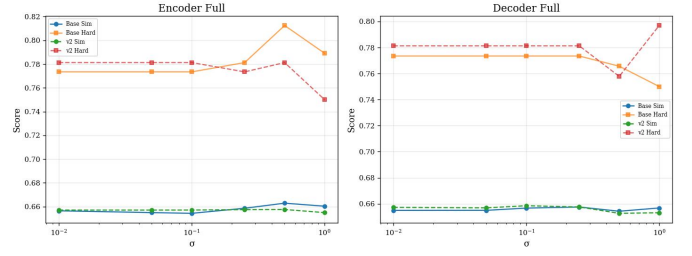


Fig. 22. Comparison between Flan-T5-Large base and MiniPLM student under Gaussian noise injection.

v1 vs v2 --- Cumulative Layer Dropping

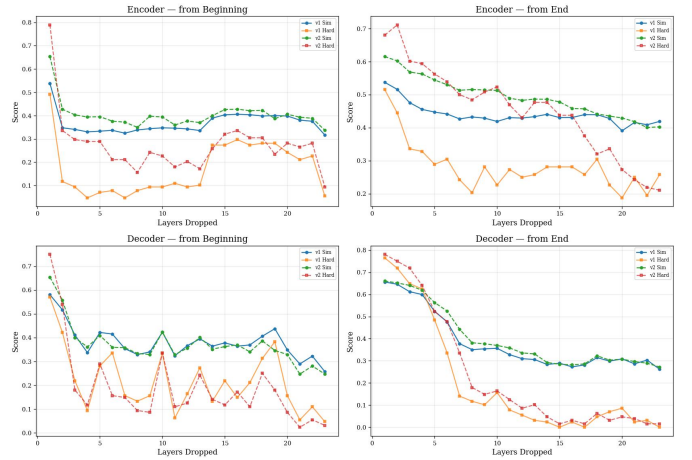


Fig. 23. Comparison between semantic distillation student v1 and MiniPLM student v2 under cumulative layer dropping.

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